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Attendance

12/8/25	Shubham Sharma	590014710
	Batch - 1	1/1
	Lab 1.	
	To implement DDL and DML	
	① What is the difference between TRUNCATE and DELETE?	
	TRUNCATE	DELETE
• Type	DDL command	DML command
• Speed	Faster (doesn't log individual Logs)	Logs each row deletion
• Rollback	Cannot be rolled back in most databases	Can be rolled back
• Space	Release storage space immediately	Doesn't release space immediately
• Locking	Locks entire table	Lock individual Rows.

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② what is the difference between TRUNCATE and DROP

		TRUNCATE	DROP
1	Purpose	Removes all data from Table	Removes entire Table structure
2	Metadata	Keeps table structure, constraints	Removes all table definitions
3	Space	Releases data space only	Releases all table space
4	Existence	Table Remains	Table is gone
5	Dependencies	Doesn't affect Views	Break dependent Objects.
3	5 tables		
4	Done		
5	Done		
6	Done		
7	Done		

Walkthrough

ADBMS

LAB 1

LAB 1: To implement DDL and DML commands

Lab Objective: To understand the concept of designing issue related to the database with creating, populating the tables. Also familiarize students with different ways of manipulation in database.

Working Procedure:

RDBMS is acronym for Relation Database Management System. Dr. E. F. Codd first introduced the Relational Database Model in 1970. The Relational model allows data to be represented in a simple row- column. Each data field is considered as a column and each record is considered as a row. Relational Database is more or less similar to Database Management System. In relational model there is relation between their data elements. Data is stored in tables. Tables have columns, rows and names. Tables can be related to each other if each has a column with a common type of information. The most famous RDBMS packages are Oracle, Sybase and Informix. Simple example of Relational model is as follows :

Degree of Relationship

- One to One (1:1): *Student Has Roll No.* One student has only one Roll no. For one occurrence of the first entity, there can be, at the most one related occurrence of the second entity, and vice-versa.
- One to Many or Many to One (1:M / M: 1): *Course Contains Students.* As per the Institutions Norm, One student can enroll in one course at a time however, in one course, there can be more than one student. For one occurrence of the first entity there can exist many related occurrences of the second entity and for every occurrence of the second entity there exists only one associated occurrence of the first.
- Many to Many (M: M): *Students Appears Tests.* The major disadvantage of the relational model is that a clear-cut interface cannot be determined. Reusability of a structure is not possible. The Relational Database now accepted model on which major database system are built.

The Degree of Relationship indicates the link between two entities for a specified occurrence of each.

E. F. Codd Rules

1. The Information Rule: All information must be store in table as data values.
2. The Rule of Guaranteed Access: Every item in a table must be logically addressable with the help of a table name.
3. The Systematic Treatment of Null Values: The RDBMS must be taken care of null values to represent missing or inapplicable information.

4. The Database Description Rule: A description of database is maintained using the same logical structures with which data was defined by the RDBMS.
5. Comprehensive Data Sub Language: According to the rule the system must support data definition, view definition, data manipulation, integrity constraints, authorization and transaction to represent missing or management operations.
6. The View Updating Rule: All views that are theoretically updateable are also updateable by the system.
7. The Insert and Update Rule: This rule indicates that all the data manipulation commands must be on sets of rows having a relation rather than on a single row.
8. The Physical Independence Rule: Application programs must remain unimpaired when any changes are made in storage representation or access methods.
9. The Logical Data Independence Rule: The changes that are made should not affect the user's ability to work with the data. The change can be splitting table into many more tables.
10. The Integrity Independence Rule: The integrity constraints should store in the system catalog or in the database.
11. The Distribution Rule: The system must be access or manipulate the data that is distributed in other systems.
12. The Non-subversion Rule: If a RDBMS supports a lower level language then it should not bypass any integrity constraints defined in the higher level.

Efficient And Effective Solutions

DBMS product that provides efficient and effective solutions for major database features. This includes:

1. Large databases and space management control
2. Many concurrent database users
3. High transaction processing performance
4. High availability
5. Controlled availability
6. Industry accepted standards
7. Manageable security
8. Database enforced integrity
9. Client/Server environment
10. Distributed database systems
11. Portability
12. Compatibility
13. Connectivity

Question:**Real World Scenario ROADWAY TRAVELS**

“Roadway Travels” is in business since 1977 with several buses connecting different places in India. Its main office is located in Dehradun. The company wants to computerize its operations in the following areas:

- Reservations
- Ticketing
- Cancellations

Reservations: Reservations are handled directly by booking office. Reservations can be made 60 days in advance in either cash or credit. In case the ticket is not available, a waitlisted ticket is issued to the customer. This ticket is confirmed against the cancellation.

Cancellation and Modification: Cancellations are also directly handed out at the booking office. Cancellation charges will be charged. Wait listed tickets that do not get confirmed are fully refunded.

AIM IS TO REPRESENT ALL ENTITIES IN A TABULAR FASHION**What is SQL?**

Oracle was the first company to release a product that used the English-based Structured Query Language or SQL. This language allows end users to manipulate information of table(primary database object). To use SQL, you do not require any programming experience. SQL is a standard language common to all relational databases. SQL is database language used for storing and retrieving data from the database. Most Relational Database Management Systems provide extension to SQL to make it easier for application developers. A table is a primary object of database used to store data. It stores data in form of rows and columns. To communicate with Oracle, SQL supports the following categories of commands:

1. Data Definition Language(DDL): Create, Alter, Drop and Truncate
2. Data Manipulation Language (DML): Insert, Update, Delete and Select
3. Transaction Control Language (TCL): Commit, Rollback and Save point
4. Data Control Language (DCL): Grant and Revoke

DATA TYPE DESCRIPTION

1. Char(Size): It Stores fixed-length character data to store alphanumeric values, with a maximum size of 2000 bytes. Default and minimum size are 1 byte.
 2. Varchar2(Size): It stores variable-length character data to store alphanumeric values, with maximum size of 4000 bytes.
 3. char(Size): It stores fixed-length character data of length size characters or bytes, depending on the choice of national character set. Maximum size if determined by the number of bytes required storing each character with an upper limit of 2000 bytes. Default and minimum size is 1 character or 1 byte, depending on the character set.
-

4. Nvarchar2(Size): It stores variable-length character string having maximum length size characters or bytes, depending on the choice of national character set. Maximum size is determined by the number of bytes required to store each character, with an upper limit of 4000 bytes.
5. Long: It stores variable-length character data up to 2GB(Gigabytes). Its length would be restricted based on memory space available in the computer.
6. Number [p, s]: Number having precision p and scale s. The precision p indicates total number of digits varies from 1 to 38. The scale s indicates number of digits in fraction part varies from -84 to 127.
7. Date: It stores dates from January 1, 4712 B.C. to December 31, 4712 A.D. Oracle predefined format of Date data type is DD-MON-YYYY.
8. Raw(Size) Stores binary data of length size. Maximum size is 2000 bytes. One must have to specify size with RAW type data, because by default it does not specify any size.
9. Long Raw: It stores binary data of variable length up to 2GB(Gigabytes).
10. LOBS – LARGE OBJECTS: LOB is used to store unstructured information such as sound and video clips, pictures up to 4 GB size.
11. CLOB A Character Large Object containing fixed-width multi-byte characters. Varying-width character sets are not supported. Maximum size is 4GB.
12. NCLOB A National Character Large Object containing fixed-width multi-byte characters. Varying-width character sets are not supported. Maximum size is 4GB. Store national character set data.
13. BLOB To store a Binary Large Object such as graphics, video clips and sound files. Maximum size is 4GB.
14. BFILE Contains a locator to a large Binary File stored outside the database. Enables byte stream I/O access to external LOBs residing on the database server. Maximum size is 4GB. Apart from oracle internal data types, user can create their own data type, which is used in database and other database object. We will discuss it in the later part.

STRUCTURE OF TABLES

The following are tabular representation of the above entities and relationships:

Bus		
COLOUMN NAME	DATA TYPE	CONSTRAINT
Bus No	varchar2(10)	Primary Key
Source	varchar2(20)	
Destination	varchar2(20)	
Couch Type	varchar2(20)	

Reservation		
COLOUMN NAME	DATA TYPE	CONSTRAINT
PNR_No	number(9)	Primary Key
Journey date	Date	
No-of-seats	integer(8)	
Address	varchar2(50)	
Contact_No	Number(9)	Should be equal to 10 numbers and not allow other than numeric
Bus_No	Varchar2(10)	Foreign key
Seat_No	Number	

Ticket		
COLOUMN NAME	DATA TYPE	CONSTRAINT
Ticket_No	number(9)	Primary Key
Journey-date	Date	
Age	int(4)	
Sex	char(10)	
Source	Varchar2(10)	
Destination	Varchar2(10)	
Dep-Time	Varchar2(10)	
Bus_No	Number2(10)	

Passenger		
COLOUMN NAME	DATA TYPE	CONSTRAINT
PNR_No	number(9)	Primary Key
Ticket_No	Number(9)	Foreign key
Name	Varchar2(15)	
Age	int(4)	
Sex	char(10)	(Male/Female)
Contact_No	Number(9)	Should be equal to 10 numbers and not allow other than numeric

Cancellation		
COLOUMN NAME	DATA TYPE	CONSTRAINT
PNR_No	number(9)	Primary Key
Journey-date	Number(9)	Foreign key
Seat_No	Varchar2(15)	
Contact_No	Number(9)	Should be equal to 10 numbers and not allow other than numeric

Practicing DDL Commands

Data Definition Language: The data definition language is used to create an object, alter the structure of an object and also drop already created object. The Data Definition Languages used for table definition can be classified into following:

Create table command

Create Table <table_name> (column1 datatype(size), column2 datatype(size), column(n) datatype(size));

Where, table_name is a name of the table and (column1, column2, ..., column n) is a name of the column available in table. Each column is separated by comma.

Desc command

Describe command is external command of Oracle. The describe command is used to view the structure of a table as follows.

Desc <table name>

Alter table command

Once Simple Table is created, if there is a need to change the structure of a table at that time alter command is used. It is used when a user wants to add a new column or change the width of datatype or datatype itself or to add or drop integrity constraints or column.

Alter table <table_name> add(column1 datatype, column2 datatype);

Add option is used with alter table_ when you want to add a new column in existing table. If you want to Add more than one column then just write column name, data type and size in brackets. As usual Comma sign separates each column. For Example, suppose you want to add column comm in emp_master, then you have to perform the following command.

ALTER TABLE Persons **ALTER COLUMN** DateOfBirth year;

Truncate table command

TRUNCATE TABLE command is used to empty a table. This command is a sequence of DROP TABLE and CREATE TABLE statements and requires the DROP privilege.

TRUNCATE TABLE table_name;

Drop table command

DROP TABLE statement is a Data Definition Language (DDL) command that is used to remove a table's definition, and its data, indexes, triggers, constraints and permission specifications (if any).

DROP TABLE table_name;

Data Manipulation Commands

Insert

INSERT INTO Statement is used to add new rows of data into a table in the database. Almost all the RDBMS provide this SQL query to add the records in database tables.

INSERT INTO TABLE_NAME (column1, column2...columnN) VALUES (value1, value2...valueN);

INSERT INTO CUSTOMERS (ID,NAME,AGE,ADDRESS,SALARY) VALUES (1, 'Ramesh', 32, 'Ahmedabad', 2000.00);

Update

UPDATE statement makes use of locks on each row while modifying them in a table, and once the row is modified, the lock is released. Therefore, it can either make changes to a single row or multiple rows with a single query.

***UPDATE table_name
SET column1 = value1, column2 = value2,..., columnN = valueN
WHERE [condition];***

***UPDATE table_name
SET column_name1 = new_value, column_name2 = new_value...
WHERE condition(s)***

Delete

DELETE TABLE statement is used to delete the existing records from a table in a database. If you wish to delete only the specific number of rows from the table, you can use the WHERE clause with the DELETE statement. If you omit the WHERE clause, all rows in the table will be deleted. The SQL DELETE statement operates on a single table at a time.

Command

DELETE FROM table_name;

DELETE FROM table_name WHERE condition;

Example

DELETE FROM CUSTOMERS WHERE NAME='Hardik';

Select

SELECT Statement is used to fetch the data from a database table which returns this data in the form of a table. These tables are called result-sets.

SELECT column1, column2, columnN FROM table_name;

*SELECT * FROM table_name;*

Lab Performance Questions:

- 1 What is the difference between TRUNCATE and DELETE?
- 2 What is the difference between TRUNCATE and DROP?
- 3 Create all the given table.
- 4 Add a new column to the existing relation.
- 5 Change the datatype of the table from char to varchar2.
- 6 Change the name of column/field.
- 7 Modify the column width of all the table

Output

Q3 Create all the given table.

The screenshot shows the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Shubham Sharma' are on the right. The 'SQL Commands' tab is active, showing a schema dropdown set to 'WKSP_KTGDBMS255'. The language is 'SQL' and rows are set to '10'. The command area contains the following SQL statement:

```
1 Create Table Bus (Bus_No varchar2(10) PRIMARY KEY, Source varchar2(20), Destination varchar2(20), Couch_Type varchar2(20));
2
```

The 'Results' tab is selected, displaying the message 'Table created.' and a duration of '0.04 seconds'. The footer shows the user 'ktg562s@gmail.com', the database 'ktg_dbms_25_5', and the version 'Oracle APEX 24.2.6'.

The screenshot shows the APEX SQL Workshop interface with the same top navigation bar. The 'SQL Commands' tab is active, with the schema dropdown set to 'WKSP_KTGDBMS255'. The language is 'SQL' and rows are set to '10'. The command area contains the following SQL statement:

```
1 Create Table Reservation (
2   PNR_No number(9) PRIMARY KEY,
3   Journey_Date date,
4   No_ofseats number(8),
5   Address varchar2(50),
6   Contact_No varchar2(10) check (Length(Contact_No) = 10),
7   Bus_No varchar2(10),
8   Seat_No number,
9   FOREIGN KEY (Bus_No) REFERENCES Bus(Bus_No)
10 );
11
```

The 'Results' tab is selected, displaying the message 'Table created.' and a duration of '0.05 seconds'. The footer shows the user 'ktg562s@gmail.com', the database 'ktg_dbms_25_5', and the version 'Oracle APEX 24.2.6'.

The screenshot shows the APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. A search bar and user profile 'Shubham Sharma' are on the right. The 'SQL Commands' section is active, showing a schema of 'WKSP_KTGDBMS255'. The language is set to 'SQL' and the number of rows is '10'. The SQL command entered is:
1 Create Table Cancellation (
2 PNR_No number(9) Primary Key,
3 Journey_date number(9),
4 Seat_No varchar2(15),
5 Contact_No number(10) Check (Length(Contact_No) = 10)
6);
7
The 'Results' tab at the bottom shows the message 'Table created.' and a duration of '0.04 seconds'.

```
1 Create Table Cancellation (  
2 PNR_No number(9) Primary Key,  
3 Journey_date number(9),  
4 Seat_No varchar2(15),  
5 Contact_No number(10) Check (Length(Contact_No) = 10)  
6 );  
7
```

Table created.
0.04 seconds

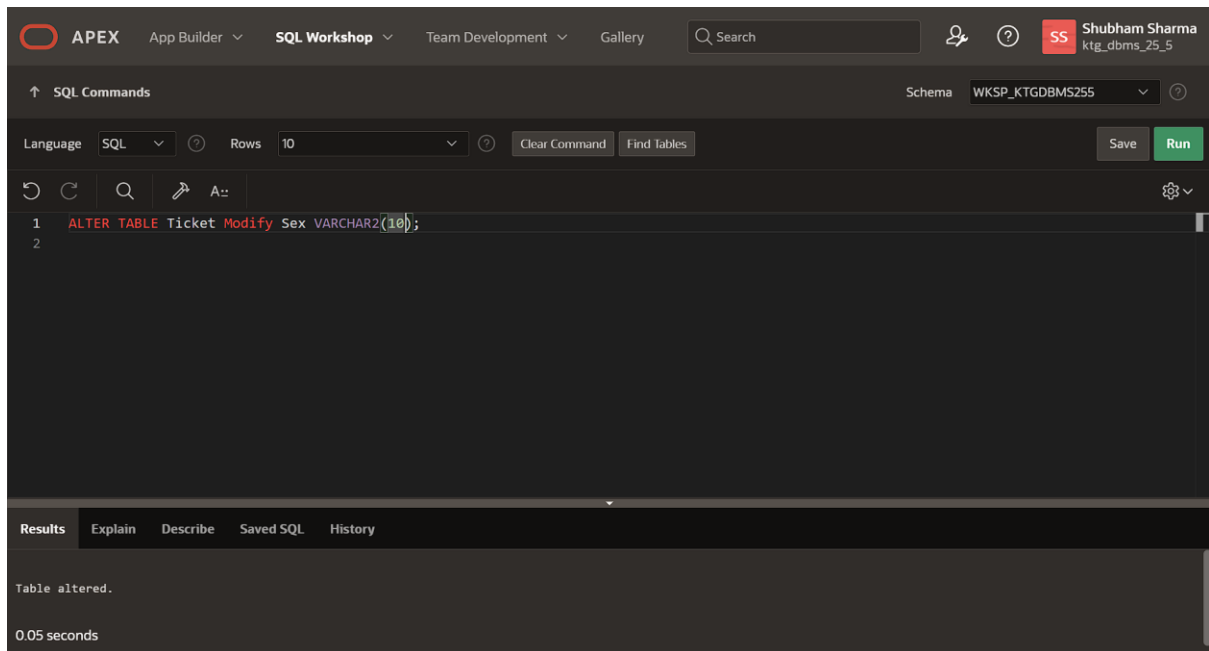
Q4: Add a new column to the existing relation.

The screenshot shows the APEX SQL Workshop interface. The top navigation bar is the same as the previous image. The 'SQL Commands' section is active, showing a schema of 'WKSP_KTGDBMS255'. The language is set to 'SQL' and the number of rows is '10'. The SQL command entered is:
1 ALTER TABLE Bus ADD Bus_Status VARCHAR2(15);
2
The 'Results' tab at the bottom shows the message 'Table altered.' and a duration of '0.05 seconds'.

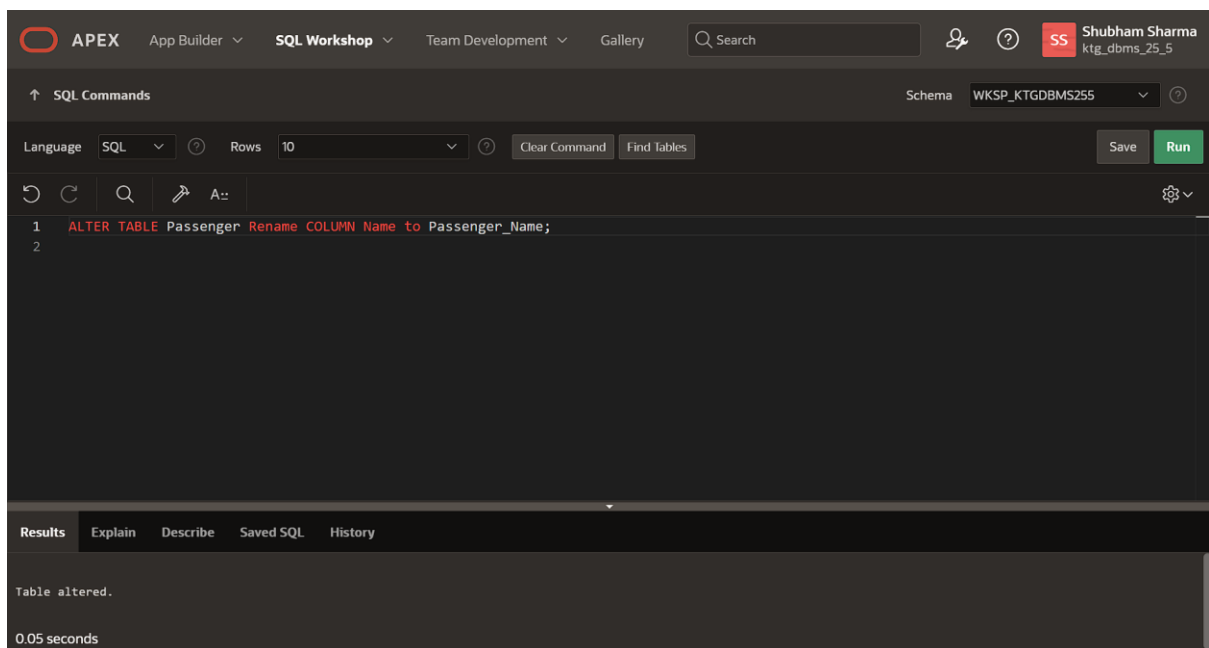
```
1 ALTER TABLE Bus ADD Bus_Status VARCHAR2(15);  
2
```

Table altered.
0.05 seconds

Q5: Change the datatype of the table from char to varchar2



Q6: Change the name of column/field.



Q7: Modify the column width of all the table

APEX

App Builder

SQL Workshop

Team Development

Gallery

Q Search

SS

Shubham Sharma

ktg_dbms_25_5

↑ SQL Commands

Schema WKSP_KTGDBMS255

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

↶ ↷ 🔍 ↵ A::

1 ALTER TABLE Bus MODIFY Source VARCHAR2(30);

Results

Explain

Describe

Saved SQL

History

Table altered.

0.05 seconds

APEX

App Builder

SQL Workshop

Team Development

Gallery

Q Search

SS

Shubham Sharma

ktg_dbms_25_5

↑ SQL Commands

Schema WKSP_KTGDBMS255

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

↶ ↷ 🔍 ↵ A::

1 ALTER TABLE Bus MODIFY Destination VARCHAR2(30);

Results

Explain

Describe

Saved SQL

History

Table altered.

0.05 seconds

APEX

App Builder

SQL Workshop

Team Development

Gallery

Q Search

?

SS Shubham Sharma

ktg_dbms_25_5

↑ SQL Commands

Schema WKSP_KTGDBMS255

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

↶ ↷ 🔍 📌 A:: ⚙️

1 ALTER TABLE Reservation MODIFY No_ofseats number(10);

Results Explain Describe Saved SQL History

Table altered.

0.04 seconds

APEX

App Builder

SQL Workshop

Team Development

Gallery

Q Search

?

SS Shubham Sharma

ktg_dbms_25_5

↑ SQL Commands

Schema WKSP_KTGDBMS255

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

↶ ↷ 🔍 📌 A:: ⚙️

1 ALTER TABLE Ticket MODIFY Age number(10);

Results Explain Describe Saved SQL History

Table altered.

0.04 seconds

APEX

App Builder

SQL Workshop

Team Development

Gallery

Q Search

?

SS Shubham Sharma

ktg_dbms_25_5

↑ SQL Commands

Schema WKSP_KTGDBMS255

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

↶ ↷ 🔍 📌 A:: ⚙️

1 Alter table Ticket Modify Dep_Time varchar2(30);

Results Explain Describe Saved SQL History

Table altered.

0.05 seconds

APEX

App Builder

SQL Workshop

Team Development

Gallery

Q Search

SS

Shubham Sharma

ktg_dbms_25_5

↑ SQL Commands

Schema WKSP_KTGDBMS255

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

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1

Insert Into Ticket (Ticket_No, Journey_Date, Age, Sex, Source, Destination, Dep_Time, Bus_No)

2

Values (235478912, TO_DATE('28-07-2025','DD-MM-YYYY'), 18, 'Male', 'Vienna', 'Warsaw', '10:30', 246);

3

4

Results

Explain

Describe

Saved SQL

History

1 row(s) inserted.

0.03 seconds

APEX

App Builder

SQL Workshop

Team Development

Gallery

Q Search

SS

Shubham Sharma

ktg_dbms_25_5

↑ SQL Commands

Schema WKSP_KTGDBMS255

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

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1

Select * From Ticket;

2

Results

Explain

Describe

Saved SQL

History

TICKET_NO	JOURNEY_DATE	AGE	SEX	SOURCE	DESTINATION	DEP_TIME	BUS_NO
235478912	7/28/2025	18	Male	Vienna	Warsaw	10:30	246

1 rows returned in 0.04 seconds

Download